

down the size of the device from 100 microns by 100 microns. This will help them in devising silicon photonic chips to direct light waves in different ways and to pack millions of beamsplitters on an individual chip.

Smartglasses that can read emotions

Microsoft has been awarded a patent for smartglasses that would allow users to detect and interpret emotions of people within their field-of-vision. Users would be able to determine whom to analyse, whether an individual or a group, without their knowing.

Sensors, including depth cameras and a microphone mounted on the nose bridge, will pick up visual and audio information from a subject. This would be processed for things like subtle variations in speech rhythm and amplitude, choice of words, type and speed of gestures, eye focus and body posture.

This information will then be beamed to Microsoft's databases and an emotional determination will be relayed back to the user through the smartglasses.

NASA 3D prints copper rocket engine part

Engineers at NASA have developed the first full-scale, 3D-printed copper rocket engine part, a combustion chamber liner that operates at extreme temperatures and pressures.



NASA 3D prints copper rocket engine part

Numerous complex parts made of many different materials are assembled to make engines that provide the thrust that powers the rockets.

Additive manufacturing has the potential to reduce the time and cost of making rocket parts like the copper liner found in rocket combustion chambers where super-cold propellants are mixed and heated to extreme temperatures needed to send rockets to space.

A selective laser melting machine in Marshall Space Flight Centre's Materials and Processing Laboratory fused 8255 layers of copper powder to make the

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Scilab-Arduino Workshop

July '15 03 July '15 04



Arduino is an open source microcontroller board and an electronic prototyping platform, popular in industry. Scilab is an open source, user friendly, state of the art, computational engine. This workshop is devoted to the control of an Arduino board from Scilab. The workshop kit will include an Arduino Uno board, a shield containing sensors and actuators, and necessary documentation. Participants will learn to perform embedded system experiments on the Arduino board using Scilab code and also through the GUI based simulation environment, Xcos.

The following experiments will be performed:

- LED Blink
- RGB LEDs
- Light intensity measurement
- DC motor control
- Pushbutton
- Relay
- Potentiometer
- Temperature sensor
- Using different sensors
- Receive data through any modbus compatible device
- Demos on some industrial applications
- ... and more



LH 101
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Register here:

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